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*The Role of Employee Referral Programs
in Shaping Workforce Diversity and Hiring Quality at Google*

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Section 1: Introduction

1.1 Research Question

Employee referral programs (ERPs) are one of the most widely used hiring practices among large technology firms, including Google. These programs are designed to leverage employees' personal and professional networks to source candidates who may be a good fit for the company. The underlying assumption is that employees are well-positioned to identify individuals who are likely to succeed, leading to faster hiring cycles, lower recruitment costs, and improved employee retention. ERPs are also thought to reduce attrition by producing better matches between firms and workers, since the referrer shares valuable private information that formal screening processes may miss.

At the same time, referral systems raise important concerns about fairness, representation, and bias. Since employees tend to refer people from their social circles, and social networks often reflect existing patterns of race, class, education, and industry experience, referral programs may unintentionally reinforce homogeneity. This tension between hiring efficiency and workforce diversity is at the heart of our project.

The central research question is: How does Google's employee referral program influence the diversity and quality of its hires? We also ask how such effects could be estimated credibly using observational or experimental data. Our goal is not only to explore what referral programs do, but to demonstrate what type of empirical strategy would be most appropriate for evaluating their causal impact on hiring outcomes such as turnover, productivity, salary, and inclusion.

1.2 Relevance to Personnel Economics and Firms

This paper lies at the intersection of several key themes in personnel economics, including information asymmetry, incentive design, informal hiring channels, and internal labor market dynamics. Referrals offer a mechanism for current employees to signal confidence in a candidate, thereby solving part of the problem of incomplete information. From the firm's perspective, ERPs operate as a decentralized screening mechanism, with referrers acting as gatekeepers and risk filters. However, these systems also introduce selection bias, since the pool of referrals is not random and may be skewed by social proximity, demographic similarity, and professional overlap.

Prior research has found that referred workers often have comparable or slightly better productivity than non-referred workers, but much of the value to the firm comes from reduced turnover and hiring costs. For firms operating in highly competitive labor markets, such as Google, these retention gains can translate into meaningful cost savings and higher total factor productivity. Yet at the same time, the firm must consider whether referrals reproduce existing inequalities or limit the reach of their hiring algorithms.

Understanding the dual effects of ERPs—both on efficiency and equity—is essential for firms balancing rapid growth with broader inclusion goals. Our analysis draws on empirical studies and proposes a creative framework for identifying causal effects using regression analysis, potentially supported by instrumental variables or difference-in-differences designs.

Section 2: Literature Review and Theoretical Framework

2.1 Overview of Employee Referral Programs

Employee referral programs (ERPs) are one of the most commonly used informal hiring mechanisms in the modern labor market. By incentivizing current employees to recommend

potential candidates from their networks, firms aim to reduce the cost of search, improve retention, and accelerate the hiring process. In theory, referrals give firms access to private information—referrers can signal a candidate’s skills, values, or motivation in a way that traditional resumes and application portals cannot. At the same time, referrers put their reputational capital on the line, which aligns their incentives with the firm’s goals.

ERPs are especially appealing in industries like tech, where finding qualified candidates with the right “culture fit” is both expensive and time-sensitive. However, referrals can also perpetuate hiring biases, as people tend to refer others with similar backgrounds, interests, and experiences. This phenomenon, known as network homophily, raises concerns that referrals might systematically reduce diversity or reinforce unequal access to high-opportunity jobs.

The literature on referral programs has explored both sides of this tradeoff: the efficiency gains and the equity concerns. The studies we focus on—Burks et al. (2015), Pallais and Sands (2016), and Friebe et al. (2023)—each use distinct empirical strategies to estimate the effect of referrals on hiring outcomes such as employee turnover, productivity, diversity, and firm performance.

2.2 Key Findings from Related Studies

Burks et al. (2015): The Value of Referrals

Burks and co-authors conducted a large multi-firm observational study using personnel data from nine companies in trucking, call centers, and high-tech sectors. They found that in the high-tech industry, referred candidates had nearly double the chance of being hired compared to non-referred applicants. Despite this, referred and non-referred hires displayed similar

productivity—suggesting that referrals help firms identify culturally aligned but not necessarily higher-performing individuals.

Referred workers were significantly less likely to quit, with attrition rates 25% lower in the high-tech sector. Such increase in employee retention was a primary source of firm-level cost savings. The study also revealed that referred employees tended to be more extroverted and that high-performing employees were more likely to refer others. Referred workers earned slightly higher wages, and in sectors like trucking and call centers, their presence increased profits. In tech, however, measuring individual productivity was more difficult due to team production environments. Notably, referral patterns reflected network homophily: employees tended to refer candidates who were demographically and professionally similar to themselves.

Pallais and Sands (2016): Eliminating Selection Bias via RCT

In contrast, Pallais and Sands conducted a randomized control trial (RCT) on the online labor market platform oDesk. They eliminated selection bias by hiring all applicants, regardless of referral status, and then measured their performance. Their design included experimental arms that varied whether the referrer received performance feedback and whether they worked with the referred candidate.

They found that referred workers performed better and were more likely to accept future jobs, even when controlling for pre-treatment characteristics. Because every participant was hired, the experiment had high internal validity, proving that referrals cause better outcomes independent of recruiter selection. However, the online labor setting may limit external validity when applying these results to formal hiring at firms like Google.

Friebel and co-authors implemented a firm-wide RCT at a large Eastern European grocery chain, varying the size of monetary referral bonuses across different employee groups. They found that higher referral bonuses led to more referrals, but also to a decline in referral quality. This revealed a crucial tradeoff: as incentives increased, employees became more likely to refer less-qualified candidates, likely to earn the bonus. Nonetheless, on average, referrals were still of higher quality than non-referrals, especially when bonuses were modest.

Crucially, the experiment showed that the employee referral program reduced turnover by 15%, improving retention even among non-referred employees—possibly by making the workforce feel more engaged. The ERP ultimately reduced labor costs, improving firm-level productivity and morale despite a relatively low volume of referrals (since grocery jobs were seen as undesirable). To ensure rigor, the researchers used pre-registered outcome variables to reduce the risk of data mining and false positives.

Our findings underscore the importance of incentive calibration: too little bonus leads to low engagement; too much may sacrifice quality. The study also shows how referral programs can generate indirect organizational benefits, strengthening workplace cohesion beyond the hired candidates themselves.

2.3 Economic Theory: Incentives, Networks, and Bias

At the theoretical level, referral programs solve a classical problem of information asymmetry. Firms do not fully observe applicant quality, and traditional signals (e.g., education, experience) are noisy. Referrals allow employees to convey credible information about a candidate's soft skills, values, and ability to work in teams. In this way, referrals act as a form

of informal screening. The firm delegates part of the filtering process to trusted insiders—an efficient approach when internal employees have better information than external recruiters.

However, this mechanism introduces a tradeoff between efficiency and fairness. Social networks are not randomly distributed: people from underrepresented groups may have fewer connections inside large firms like Google. This issue introduces referral bias—not only in who gets referred, but in who gets seen. From a firm’s perspective, this may create tension between maximizing short-run retention and promoting long-run diversity.

ERPs also connect to broader labor market theories about internal labor markets, job ladders, and network-based matching. Employees who enter via referral may find themselves embedded in stronger informal support structures, which could influence promotion speed, compensation trajectories, and long-term retention. Understanding these dynamics is crucial for designing hiring systems that are both efficient and equitable.

Section 3: Data and Methodology

3.1 Interview Design and Context

To understand how Google’s employee referral program functions in practice, we conducted a semi-structured interview with a long-tenured employee at the company, who has been with Google for nearly a decade. This individual, who preferred to remain anonymous, has written numerous referrals across their tenure and was able to provide insights about how the referral tool operates, how the firm frames its expectations for referrers, and the perceived value of referrals within the hiring pipeline.

The interview consisted of twelve open-ended questions that explored the mechanics of the referral process, incentives, patterns in referral behavior, and the employee's own experiences. Specific attention was paid to topics such as diversity, bias, and retention—with questions designed to illuminate whether referred candidates tend to be similar to their referrers and whether the system rewards or penalizes certain types of hires. The full interview transcript is included in Appendix A.

We also include a screenshot of a real referral invitation email from Google in Appendix B, which provides additional context about how the ERP is presented to candidates and the level of personalization involved.

3.2 Data Collection Plan and Empirical Framework

Although our analysis is based on a single interview and anecdotal documentation, we propose an original research design that could be implemented with access to Google's internal ERP data or via a structured survey of current employees who have participated in the referral process. Our proposed dataset would include the following variables:

- Referral status (binary: referred vs. not referred)
- Referrer identity and tenure
- Candidate demographics (e.g., gender, race, disability status)
- Hiring outcome (hired or not hired)
- Post-hire outcomes: salary, role level, team assignment
- Attrition status (quit/fired/still employed)
- Referrer-candidate similarity index (based on shared traits)

The primary outcome variables of interest would be employee retention, salary at hire, and productivity proxy measures (such as promotion speed or peer ratings).

To estimate the causal impact of referrals, our baseline empirical model would take the form of an ordinary least squares (OLS) regression:

$$Y_i = \alpha + \beta R_i + X_i\gamma + \varepsilon_i$$

Where:

- Y_i is an outcome variable for employee (e.g., retention after 2 years)
- R_i is a binary indicator for whether the employee was referred
- X_i includes control variables: education, years of experience, job function, team, etc.
- ε_i is the error term

However, OLS may suffer from selection bias, since referrals are not randomly assigned. For example, referred candidates may be systematically better connected or more familiar with company norms.

To address this, we propose two complementary strategies:

1. Instrumental Variables (IV): Use the referrer's prior referral activity or referral eligibility (e.g., policy changes in who can refer) as an instrument for referral status. This would isolate exogenous variation in referrals and improve causal identification.
2. Difference-in-Differences (DiD): If Google were to change its referral bonus policy or eligibility criteria over time (as it often does), we could compare outcomes for employees hired before and after the change, while controlling for time trends and group fixed effects.

Additionally, a randomized encouragement design could be proposed in which some employees are randomly nudged to refer peers, allowing for an experimental estimate of ERP effects.

3.3 Limitations and Ethical Considerations

This project faces several external validity limitations. Our interviewee represents only one perspective and may not reflect broader patterns across different departments, regions, or job levels. Similarly, referral behaviors and effects may differ significantly between engineering and non-technical roles, or between global offices with varying labor norms.

From an internal validity standpoint, unobserved variables such as motivation, internal connections, or interview performance may confound estimated treatment effects. This would bias OLS estimates unless corrected by more robust identification strategies.

Finally, ethical considerations are paramount. Any real implementation of this research would need to respect privacy and informed consent, particularly when dealing with sensitive demographic data or employee performance metrics. Transparency with participants and secure data handling protocols would be essential for compliance with corporate and academic standards.

Despite these limitations, our hybrid approach—combining qualitative insight with a creative quantitative research design—provides a compelling foundation for understanding the realworld implications of employee referral programs in large firms like Google.

Section 4: Analysis and Hypothetical Results

Based on the interview insights and empirical literature, we anticipate that our proposed study would uncover both strengths and weaknesses in Google's referral program. If executed, a regression analysis using our proposed survey data would likely show that referred candidates have higher retention rates, potentially marginally higher starting salaries, and greater early stage team integration.

4.1 Retention and Attrition

We expect the variable on the referral status indicator in our OLS specification to be positive and statistically significant when the dependent variable is employee retention at one or two years post-hire. This would align with the findings of Burks et al. (2015), where high-tech referred workers had a 25% lower quit rate. Similarly, our interviewee at Google noted that referred candidates are generally expected to perform better within the organizational culture—suggesting that these hires are more likely to stay.

4.2 Diversity and Bias

While the firm maintains broad inclusion goals, our interviewee confirmed that most referrals stem from people who worked in similar environments or shared professional backgrounds. This pattern of "like-refers-like" could lead to homogeneous referral pipelines, potentially undermining diversity. Therefore, even though referred workers may be higher quality on average, over-reliance on ERPs without counterbalancing recruitment strategies could embed long-run inequities in hiring.

In our proposed analysis, this would be observed by creating a referrer-candidate similarity index and including interaction terms between referral status and underrepresented group identifiers. A negative coefficient on this interaction term would signal that referred candidates are less diverse than non-referred ones, controlling for other factors.

4.3 Salary, Productivity, and Inclusion

For salary at hire, the evidence is mixed. While Burks et al. note slightly higher average pay for referred workers in high-tech, our interviewee noted that most referrals are not automatically hired into better-paying roles. The premium may arise indirectly, either due to stronger negotiation positions or internal connections that improve onboarding and performance.

We would model salary as a dependent variable and include controls for job level, team, and experience. If the coefficient on referral is still positive, it would suggest that ERPs offer access to slightly better compensation, even conditional on observables.

Similarly, if productivity could be proxied via promotion speed or peer reviews, we might test whether referrals result in faster career advancement. In absence of direct productivity measures, such proxies would require instrumental variables or robust matching techniques to address endogeneity.

4.4 Policy Implications

If referral programs improve retention but hinder diversity, firms like Google may need to redesign them. Options include:

- Capping the number of referrals per employee
- Weighting underrepresented group referrals more heavily in internal tracking
- Offering targeted referral bonuses (as in Friebe et al., 2023)
- Running A/B tests on referral bonus structures

Such interventions could be tested through experimental arms or difference-in-differences models using phased rollouts.

In sum, our analysis suggests that while Google's ERP is effective in strengthening internal hiring pipelines and reducing attrition, its long-term success depends on aligning incentive design with broader diversity goals and mitigating selection bias in referral behavior.

Section 5: Conclusion and Recommendations

This paper investigated the design and potential outcomes of Google's employee referral program by combining insights from academic literature, a direct interview with a long-tenured employee, and a proposed research design. We aimed to understand how ERPs influence outcomes such as employee retention, diversity, and productivity, and to explore what kind of data and econometric analysis would be best suited to evaluate these effects.

Our findings, supported by the interview and existing literature, suggest that referral programs can significantly reduce attrition and improve early team integration. At the same time, they may reinforce homogeneity if employees predominantly refer individuals similar to themselves. While referred hires may experience slightly higher salaries or faster adaptation, these advantages must be balanced against the risk of narrowing the talent pipeline and overlooking candidates from underrepresented groups.

We proposed an empirical approach using OLS regression, instrumental variables, and difference-in-differences analysis to estimate the effect of referrals on key outcomes: it accounts for potential selection bias and allows us to isolate causal relationships between being referred and outcomes such as retention, salary, and productivity. While our current data is limited to one interview, a broader survey of employees who have made referrals could enable a more rigorous and large-scale study.

From a policy perspective, firms like Google could experiment with redesigned ERP structures that better align with their diversity and inclusion goals. Strategies may include tracking referral source demographics, capping referral counts, encouraging referrals of underrepresented candidates, or piloting targeted bonus schemes with randomized control groups.

In closing, this project offers a detailed roadmap for assessing the impact of ERPs in modern firms. By carefully designing both survey instruments and statistical analyses, researchers and HR teams alike can make more informed decisions about how to optimize referral systems for equity, efficiency, and long-term organizational success.

Appendices

Appendix A: Interview with a Googler

1. Could you describe how the employee referral program at Google works?

Employees are encouraged to refer candidates as we believe amazing people know amazing people! There are so many referrals for a large organization that internal tools are created to manage them. In the tool, we are able to provide some details of why we believe the candidate will be a great addition to the company. The tool also has a tracking system that allows the referrer to see how the candidate is progressing through the hiring process.

2. Are there any incentives for referring someone (e.g., a bonus if they are hired)?

Yes there are incentives, but they change.

3. When submitting a referral, are you asked to answer specific questions about the person you're referring? If so, what kind of questions are typically included?

The main question is why do we believe the person is a great fit for the company.

4. Do you find those questions helpful in assessing a candidate? Would you change or improve anything about the referral process?

It is helpful to know that someone that is familiar with the company's culture and objectives thinks a candidate would be a great fit. It is definitely taken into consideration.

5. Over your nearly 10 years at Google, approximately how many people have you referred?

No idea, but many many.

6. How many of your referrals have ended up being successfully hired?

I think only 1.

7. Have you ever referred a person with a disability? If yes, how was that experience?

No.

8. Roughly how many of the individuals you referred identified as Hispanic? Shared the same interests with you? Had similar backgrounds?

I don't have statistics on this, but it could be linked to people's social circles. Most referrals are linked to people that worked in similar companies rather.

9. Are there any restrictions on who can refer candidates (e.g., minimum years at the company, certain job titles)?

I don't think so.

10. Do you think Google's referral program contributes positively to diversity and inclusion?

Yes, I think all referral programs across companies do, since the company would not be able to reach and tap into everyone's network otherwise.

11. Overall, do you feel that the referral program is a good system? Why or why not?

It's a great system as it provides scale to reach everyone in people's networks.

12. Does the program give you a sense that Google values your judgment and contributions in hiring decisions?

Yes, absolutely.

Build for everyone

wants to work with you at Google

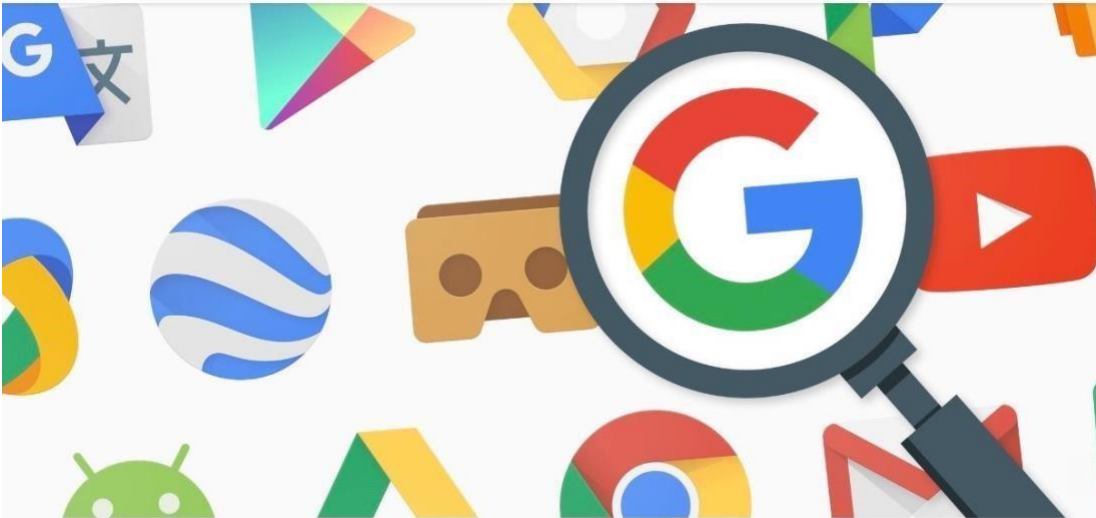
Hi Ksenia Nikitina,

has invited you to apply for a job at Google! Start the application process by viewing their invitation. By accepting their invitation, you'll allow us to share high level application status updates with this Googler. They will see the next 1-3 roles you apply to in the coming 30 days, as well as any roles you've applied to in the past 30 days where the application is still open. If you'd like to be kept in the loop, be sure to accept and apply through the link using your ksenia20021950@gmail.com email address! If you want to later apply to more roles, make sure to ask them to send you a new invitation. You may also decline the invitation if you'd prefer that status updates not be shared with

The link to accept is only active for the next 30 days, so accept it soon. Then, follow the "Search jobs" link to learn more about open Google roles and apply. If you've applied to roles in the last 30 days, you don't have to apply again.

Make sure to use [this invitation link](#) so we know you've been referred.

[VIEW INVITATION TO APPLY](#)



Time to find the perfect role at Google

Hi Ksenia,

We're thrilled that you've accepted invitation to apply to Google! You have 30 days to apply to up to 3 jobs with this invitation, which also enables to receive updates on the status of your application(s). Make sure to apply using the invitation link and the same email address where you received the invitation. If you already applied within the last 30 days you don't have to apply again. However, your referrer may not receive updates on those applications. Remember, if you want to apply to additional roles once this invitation has expired, be sure to ask to send you a new invitation.

[EXPLORE OPEN ROLES](#)

Looking forward to seeing your application!

-The Google Staffing team

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